SQL Exercise 2: Aggregate functions & Grouping.

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Nyardaro Nathoma.

① COUNT() function.

```
SELECT COUNT (id) AS total_number_of_employees
FROM employees;
```

total_number_of_employees
10

② SUM() function

```
SELECT SUM (salary) AS total_salary,
       department
FROM employees
WHERE department = 'IT';
```

total salary	department
220 000	IT

③ AVG() function

```
SELECT AVG (salary) AS average_salary,
       department
FROM employees
WHERE department = 'HR';
```

Average_salary	department
49 500	HR

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④ MIN() and MAX() functions

```
SELECT MIN(salary) AS lowest_salary,
       MAX(salary) AS highest_salary
FROM employees;
```

lowest_salary	Highest_salary
48000	62000

⑤ GROUP BY Statement

```
SELECT SUM(salary) AS total_salary,
       department,
       COUNT(id) AS total_employees
FROM employees
GROUP BY department;
```

total_employees	department	total_salary
4	IT	220 000
2	HR	99 000
2	finance	119 000
2	Marketing	105 000

⑥ GROUP BY and COUNT()

```
SELECT COUNT(id) AS employees,
       City
FROM employees
GROUP BY city;
```

employees	City
2	New york
3	Chicago
2	Los Angeles
2	San francisco
1	Houston

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⑦. GROUP BY and ORDER BY

```
SELECT department,
       AVG(Salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY ~~DESC;~~ average_salary DESC;
```

department	average_salary
finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

⑧. HAVING Clause

```
SELECT department,
       SUM(Salary) AS total_salary
FROM employees
GROUP BY department
HAVING total_salary > 100 000;
```

department	total_Salary
IT	220 000
finance	119 000
Marketing	105 000

Exercise 2: Aggregate functions & Grouping

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9. Combining GROUP BY, HAVING and ORDER BY

```
SELECT COUNT(id) AS employees,
       city
FROM employees
GROUP BY city
HAVING employees > 1
ORDER BY employees DESC;
```

employees	City
1	Houston
2	New york
2	Los Angeles
2	San francisco
3	Chicago

10. Combining Aggregate functions.

```
SELECT department,
       AVG(salary) AS Average_Salary
FROM employees
GROUP BY department
~~WHERE average_salary~~
ORDER BY average_salary DESC
LIMIT 1;
```

department	average_salary
finance	59500